GENERATIVE ENGINE OPTIMISATION FOR FLIGHT CENTRE

TECHNICAL REPORT AND HANDOVER GUIDE

IFN736 INDUSTRY PROJECT · ASSESSMENT 4: PROJECT ARTEFACT

A controlled test of whether clearer Flight Centre web pages lead AI answer engines to cite Flight Centre more often, with the results, the evidence behind them, and a guide to using and extending the testing environment.

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ABOUT THIS DOCUMENT

Purpose	Explains what we built and tested, what we found, and how Flight Centre can use, check and extend the work.
Readers	Flight Centre's digital, SEO and content teams, developers who take over the code, and QUT assessors.
Companion documents	Test Plan and Validation Record, and the project records: README, scoring rubric, baseline lock record, decisions log and claim–evidence ledger.

PART

01

THE PROJECT AT A GLANCE

What we set out to do, what we found, and what Flight Centre receives.

> **In our controlled test, clearer Flight Centre pages lifted the share of AI answers citing Flight Centre from 40.0% to 52.4%.**

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01 EXECUTIVE SUMMARY

We tested whether changing saved Flight Centre pages alters answer-level citation inclusion when FC is already among the supplied sources. Baseline: 116/290 (40.0%). The selected page produced 143/290 (49.3%) initially and 152/290 (52.4%) on repeat. Both exploratory intervals exclude zero. Historical controls, capture differences and iterative selection limit causal interpretation; this is not a forecast of live visibility or bookings.

52.4%
of AI answers cited Flight Centre in the repeat run, up from 40.0%

+12.4
percentage points in the repeat run (+9.3 in the first run)

290
AI answers per version, across 29 travel questions

6
page-change methods, each tested on its own

The problem. AI tools such as ChatGPT search and Perplexity answer travel questions by reading web pages and naming a few of them as sources. If Flight Centre is not one of the named sources, customers do not see it in the answer, even when the AI has read its page.

What we built. We could not change Flight Centre's pages inside a live AI engine, so we built our own testing environment, which we call the sandbox. It holds a dated copy of real Brave Search results for Australia: 50 travel questions and 953 web pages, captured in August 2026. A working copy of an AI answer engine answers each question from those pages, and our code checks every answer for a Flight Centre citation. Between the two versions we compare, only Flight Centre's page changes. Every competitor page stays exactly the same.

What we found.

Five of six individual estimates were positive; two intervals include zero.

Structured facts had the largest content estimate. The tested rewrite declined. Individual gains do not add.

- The final page combines four changes: deal content that normally only appears after JavaScript runs, a short question-and-facts section, FAQ questions restored to the page text, and the FAQ section moved higher up the page.
- Moving Flight Centre's page to the top of the AI's list of sources raised citations by 18.3 points. Position clearly matters, but a website owner cannot control it, so it is not part of our recommendation.
- Two extra checks outside the main test were inconclusive (Section 8.3).

What we recommend. Apply the tested content pattern where appropriate: readable offers, labelled facts and FAQ questions beside their answers. Maintain current offers, appropriate structured data and clear booking or expert pathways. Monitor access, citations and commercial outcomes through normal operations. FC does not need to repeat our sandbox. Appendix C aligns these recommendations with the final leadership presentation.

02 THE ARTEFACT AT A GLANCE

Everything Flight Centre needs is delivered in two places. The public GitHub repository holds the code and a summary of the results, and the assessment ZIP holds the full evidence. The easiest place to start is the START_HERE.html page at the top level of the ZIP. Table 1 lists the five components agreed in the Artefact Agreement and where to find each one.

TABLE 1 What Flight Centre receives

COMPONENT (ARTEFACT AGREEMENT)	WHAT IT CONTAINS	WHERE TO FIND IT
1. GEO Experiment and Optimisation Software	The code that captures the search snapshot, asks the AI, changes the pages, scores the answers and checks the results. The final page and each individual method are kept as separate, frozen copies of the code. Re-running the AI step needs your own API keys.	`03_code` in the ZIP and on GitHub
2. Optimised Travel Page Demonstrator	52 improved Flight Centre pages, each beside its original version, with fingerprints that prove which version was tested.	`04_evidence/pages` in the ZIP
3. Experimental Results and Reproducibility Pack	Every saved AI answer with its numbered list of sources, all results, a results explorer you can open in a browser, and scripts that recount the results without calling any AI service.	`04_evidence` and `03_code` in the ZIP; summaries in `results` on GitHub
4. Technical Report and Handover Guide	This report, the Test Plan and Validation Record, and the project records: README, scoring rubric, baseline lock record, decisions log and claim–evidence ledger.	`02_documentation` in the ZIP; the README and claim–evidence ledger are also on GitHub
5. Final Presentation and Presenter Notes	Both labelled decks and notes. See 01_presentation.	

As set out in the Artefact Agreement, the artefact is a research prototype. It demonstrates the approach and gives Flight Centre the tools to repeat and extend the test, but it is not a production system.

Where to start.

- If you work in marketing or SEO, read Sections 1, 8 and 15.
- If you are a developer taking over the code, read Sections 5, 14 and 15, then open the `03_code` folder.
- If you are assessing the project, read Sections 2, 9 and 10 and the Test Plan and Validation Record, then open START_HERE.html.

PART

02

WHY AND HOW WE TESTED

The business problem, the testing environment we built, and how we judged the results fairly.

> **Only Flight Centre's page changes between the versions we compare. Every competitor page stays exactly the same.**

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03 BACKGROUND AND OBJECTIVES

3.1 WHY THIS MATTERS TO FLIGHT CENTRE

More and more travel research now starts with an AI answer instead of a list of links. Three recent findings show why being cited in those answers matters:

- **AI summaries reduce clicks on ordinary results.** In a Pew Research Center study, Google users clicked a traditional result in 8% of visits when an AI summary appeared, compared with 15% when it did not, and clicked a link inside the summary in just 1% of visits (Chapekis & Lieb, 2025). Increasingly, being named in the answer is what gets a brand seen.
- **AI referrals to travel sites are growing fast.** Visits to travel websites from AI tools rose 119% year on year in July 2026, and they converted almost as well as other traffic, only 1% lower (Adobe Digital Insights, 2026). These figures come mainly from the United States.
- **Flight Centre is already investing in this area.** Its online leisure sales (total transaction value) passed $1.8 billion in FY26, up 17%, and natural language search is now live across Flight Centre and Travel Associates (Flight Centre Travel Group, 2026).

3.2 WHAT WE FOUND IN SEMESTER 1

Our dated audits found gaps in sampled unbranded answers, but other query classes showed non-zero FC visibility. They do not establish universal absence. Sampled deal content depended on JavaScript. Our text-cleaning step also removed FAQ questions inside accordion buttons. The later October checks show some FAQ text is now present without JavaScript; page versions and capture dates matter.

3.3 RESEARCH QUESTION AND SUCCESS CRITERIA

This semester we set out to answer one question: when Flight Centre's page is among the sources an AI engine reads, does changing how that page presents its information make the AI more likely to cite Flight Centre? We judged the project against the criteria in Table 2.

TABLE 2 Success criteria and outcomes

CRITERION	OUTCOME
The comparison is fair: the fairness checks pass, and every pair of answers uses the same sources and competitor text	**Met.** All checks passed and all 290 final pairs matched.
The final page beats the original pages with a 95% interval above zero	**Met** in both runs.
Stretch target: 60% of answers cite Flight Centre (set on 20 September 2026 as an aspiration, not a stopping rule)	**Not met.** The best run reached 52.4%.
The headline numbers can be recounted offline without calling any AI service	**Met.** The totals of 116, 143 and 152 citing answers were reproduced.
The gain also holds for reworded questions, and when the system has to find pages itself	**Not shown** (Section 8.3).

04 OUR APPROACH AND WHY WE CHOSE IT

4.1 WHY WE BUILT A SANDBOX

A saved-source sandbox lets us hold competitor text and source order constant while changing FC content. It isolates one part of AI visibility, although historical control reuse and capture differences still limit causality. Live evaluations are also possible, but require designs that account for changing platforms and other confounders.

TABLE 3 Ways we could have tested GEO

OPTION	WHY WE DID OR DID NOT USE IT
Ask live AI engines before and after a change	Live answers can vary with query wording, platform changes and time. Repeated dated observations can monitor real visibility, but a simple before/after comparison alone does not isolate the effect of a page change.
Publish changes on the live site and watch the results	Publishing needs normal development, content and commercial review. Live monitoring can track customer outcomes. Causal evaluation would need additional controls; it is optional research, not a prerequisite for adopting the recommendations.
Buy an AI visibility monitoring tool	Visibility tools monitor published content using their own prompt panels and denominators. They can support routine reporting but cannot evaluate unpublished page substitutions inside a commercial engine.
Build a frozen sandbox (our choice)	The same question, with the same competitors, can be asked hundreds of times for the cost of API calls. The sandbox shows harm as well as gain (it caught the AI rewriting method), and it can be reused to test any new page template or method. The trade-off is that its results are relative and apply inside the sandbox.

4.2 WILL THE RESULTS STAND THE TEST OF TIME?

The specific numbers will date, but the tool and the main finding will not.

- **The numbers belong to one snapshot.** The figures of 40% and 52% come from the August 2026 snapshot and the saved model configuration. They support content improvement under these conditions, not a production uplift forecast.
- **The sandbox stays useful.** When Flight Centre's page templates change, a developer can capture a new snapshot and run a new baseline. When the AI model changes, the same frozen pages can be tested under the new model, which gives a fair comparison between models. A model change does not undo past results; it simply creates a new test condition.
- **The finding is not tied to any one model.** Every current AI answer engine reads page text and then decides what to cite. Public crawling, search indexes, third-party sources and direct app/API integrations are distinct routes to an answer.

4.3 WHY WE CHOSE THESE METHODS

Each method we tested responds to a problem we found in Semester 1 or to a result reported in published research. Table 4 also lists the alternatives we considered and why we set them aside.

TABLE 4 The methods we tested and why

METHOD	WHY WE CHOSE IT	ALTERNATIVES WE SET ASIDE
Rendered content and cleanup	Semester 1 showed that deals and fares only appear after JavaScript runs, and the major AI crawlers do not run JavaScript (Vercel, 2024).	Re-rendering competitor pages as well, which would have broken the fair comparison.
Grounded FAQ	FAQ questions were hidden inside accordions, and a question-and-answer format matches the way customers ask.	AI-written FAQs, which can invent facts; FAQ schema on its own, which Google says is not needed.
Structured facts	The GEO paper found that clear statistics and structure improve visibility (Aggarwal et al., 2024), and Flight Centre's fares and fees were scattered across the page.	Keyword stuffing, which the GEO paper found unhelpful and Google's spam policies forbid.
Schema with visible structure	Schema.org markup is the standard way to label facts for machines.	Schema with no visible change, whose effect our test could not isolate.
External facts with attribution	The GEO paper reported gains from adding cited sources and quotations.	Repeating the customer's question on the page, which would need the question in advance.
Automatic AI rewriting	Recent research proposes rewriting whole pages for AI engines.	Human rewrites of all 52 pages, which were out of scope.
Source position (a test, not a method)	Studies show that where a source sits in the AI's input can matter more than its content (Liu et al., 2024; Puerto et al., 2025).	Freshness signals, which cannot be tested on a frozen snapshot.

05 HOW THE SANDBOX WORKS

The sandbox is a seven-stage pipeline written in Python (Figure 1). It follows the same basic pattern as AI search tools: find relevant pages, give them to a language model, and ask it to answer with numbered citations (Lewis et al., 2020; Aggarwal et al., 2024). Every input is frozen and fingerprinted, and only the Flight Centre page changes between the original and improved versions. Competitor pages never pass through the page-change stage, so any change in citations can be traced back to Flight Centre's page.

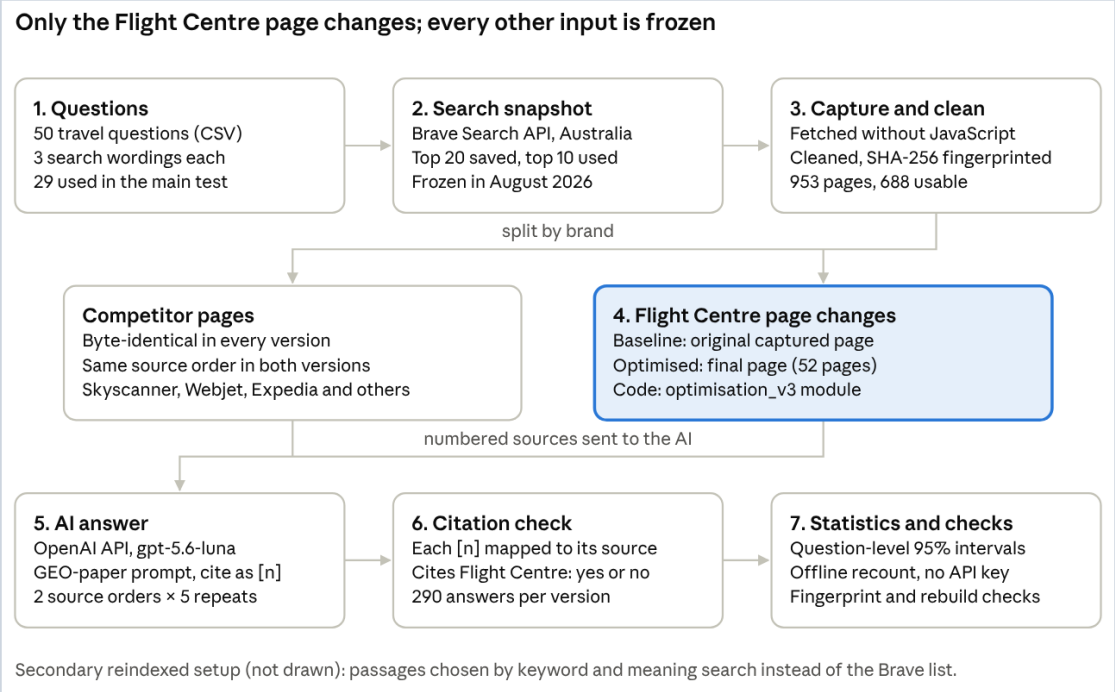


FIGURE 1 How the sandbox works. A question passes through seven stages, and only the Flight Centre page (stage 4) differs between the two versions being compared.

Four design rules are built into the code to keep the experiment honest:

- **One settings file.** Every experimental setting lives in a single file, and any change must be recorded, with its date and reason, in the decisions log.
- **A protocol lock.** Before a reportable run, the code records the model, the instruction and the settings in a fingerprinted lock file. If any setting has changed since, the run stops.
- **Write-once results.** Each run writes to a new folder, and the scripts refuse to overwrite earlier results.
- **A swappable AI provider.** The same code can run on OpenAI or Anthropic models, on a local model, or in an offline test mode that needs no API key. Only OpenAI and Anthropic runs count as reportable results.

Appendix B lists the code behind each stage.

06 HOW WE EVALUATED THE RESULTS

Every comparison keeps the questions, the sources, their order and the competitor text the same. Only the Flight Centre page changes, and we count how many of 290 answers cite Flight Centre.

6.1 TWO TEST SETUPS

We used two test setups (Table 5). Every headline result in this report comes from the main setup.

TABLE 5 The two test setups

SETUP	WHAT THE AI READS	ROLE
Main (fixed): saved source-page text, capped at 24,000 characters per source. The headline comparison uses 29 eligible flight questions with FC present in the supplied source pool.		Every headline result in this report
Secondary (reindexed): the retrieval pipeline selects five 350-word passages from the indexed usable corpus, using keyword and semantic search with at most two passages per brand. Forty questions were evaluated.		Tests whether Flight Centre gets found in the first place

6.2 WHY 290 ANSWERS, AND WHY TEN PER QUESTION

Each version contains 290 answers: 29 questions, two source orders and five repeats. Generation can vary, so repeats sample some of that variability. Both conditions use the same source order within each pair. The independent units for the bootstrap are questions, not 290 unrelated answers.

Table 6 preserves historical, assumption-dependent power estimates. These are provisional calculations, not guaranteed detection thresholds. More repeats do not replace broader query coverage. Prospective sample size should use pilot variance and the smallest effect worth detecting.

TABLE 6 Provisional power estimates

ANSWERS PER QUESTION	ANSWERS PER COMPARISON	PROVISIONAL ESTIMATE
5	290	About 12–13 points
10 (our design)	**580**	**About 9–11 points**
20	1,160	About 7–9 points
50	2,900	About 6–8 points

The estimates above do not establish that five repeats would miss a gain or that 60 questions are sufficient. Interpret each effect using its interval and design limitations. A future confirmatory study should specify its effect of interest and assumptions before calculating sample size. No new experiment is required for FC to implement content improvements.

6.3 WHAT WE MEASURED

Our main measure is **citation inclusion**: the share of answers that cite Flight Centre at least once. An answer counts if at least one of its numbered references, such as [7], points to a Flight Centre source, and several references in one answer still count only once. Citation inclusion is not the same as a brand mention, a recommendation, a click or a booking.

We also recorded two secondary measures: Flight Centre's share of all citations, and the GEO paper's position-adjusted word count. We built an AI quality judge as well, but because it was never checked against human graders, we do not use its scores.

6.4 HOW WE KNOW A CITATION POINTS TO FLIGHT CENTRE

Our check never reads the wording of the answer. Instead, it relies on the way each answer is produced:

1. For every answer, the AI receives the question and a numbered list of sources, for example [1] to [8]. That exact list, including each source's web address, brand and full text, is saved with the answer and fingerprinted.
2. The instruction, taken from the GEO paper, tells the AI to cite the sources it used by number.
3. The AI has no internet access during the test, so a reference such as [7] can only mean source 7 in that answer's list.
4. Our code looks up every reference. If any of them points to a source whose brand is Flight Centre, the answer counts as citing Flight Centre.

The scoring function is only a few lines long:

```
# 03_code/verify_results.py (standard library only, no AI calls)
def inclusion(r):
    ids = {int(x) for x in re.findall(r'\[(\d+)\]', r['answer'])}
    return int(any(0 < i <= len(r['sources'])
                   and r['sources'][i-1]['brand'] == 'flightcentre'
                   for i in ids))
```

A sentence may be supported by several sources. We count whether a valid numbered reference identifies FC in that answer, even if competitors contain the same fact. This measures citation inclusion, not unique contribution or factual support. Identical competitor text improves comparability, but generation variation, historical controls and changed capture content also affect interpretation.

The check has two limits. It does not prove that Flight Centre's text actually supports the sentence. Research on AI citations treats citing a source and being supported by it as separate questions (Gao et al., 2023), and checking support needs a human review that is still outstanding. Nor does the check show that the AI recommends Flight Centre.

6.5 STATISTICS AND FAIRNESS CHECKS

We report 95% intervals from a paired bootstrap that resamples whole questions 20,000 times (seed 20261018). The intervals are exploratory, because they are not adjusted for the many comparisons we made while developing the methods. One point needs to be stated openly. Before testing began, we pre-declared the reindexed setup as the primary measure, and the fixed comparison was declared secondary before it ran. The headline results in this report come from the fixed setup, so we report both, including the fact that the pre-declared primary measure did not move.

Four fairness checks passed before we tested any page change. Repeated retrieval returned identical passages (T1). Competitor pages were byte-for-byte identical across versions (T2). Changing Flight Centre's text did not reshuffle the competitors (T3). Rebuilding the search index produced identical passages (T4). In the final comparison, all 290 pairs of answers also matched on question, sources, source order and competitor text.

6.6 WHAT COUNTS AS A MEANINGFUL CHANGE?

There is no industry standard for a “good” increase in AI citations, so we judged each result against the noise and sensitivity of our own test:

The two same-page treatment runs differed by 3.1 percentage points. This is one observed difference, not a noise threshold.

An interval crossing zero leaves the direction uncertain; it does not prove there is no effect.

Both selected-comparison intervals exclude zero, subject to the stated exploratory limitations.

- **About 18 points** is the effect of moving Flight Centre to the top of the source list: the biggest lever we found, and one that page content cannot control.
- **20 points** would have reached the 60% stretch target, which we did not achieve.

The selected gain is a useful descriptive finding with a positive exploratory interval. It does not establish a general threshold for GEO success, a production uplift, or a causal contribution for each component. Repeated results and transparent limitations matter more than a target percentage.

PART

03

WHAT WE CHANGED AND WHAT WE FOUND

The page changes we tested, what each one did to AI citations, and real evidence from the sandbox.

Five individual estimates were positive; two were uncertain.

The tested automatic rewrite declined in this study.

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07 THE PAGE CHANGES WE TESTED

7.1 THE SIX METHODS

We built six page-change methods and one position test. Each was tested on its own against the same original pages, and each is kept as a separate copy of the code with its own unit tests. All of them follow the same four safeguards:

The selected stack uses saved FC facts; external-facts methods are separate.

- If a page layout is unfamiliar or ambiguous, they leave the page exactly as it was.
- They never look at the test questions or the answers.
- They record a fingerprint of the page before and after the change.

Table 7 summarises what each method changes and whether it is part of the final page.

TABLE 7 The methods and what they change

METHOD	WHAT IT CHANGES ON THE FLIGHT CENTRE PAGE	IN THE FINAL PAGE?
1. Rendered content and cleanup	Uses the version of the page captured after JavaScript has run, so deal cards and fares become readable text. It also tidies the order of sections and fixes prices broken up by stray spaces.	Yes
2. Grounded FAQ	Adds a question and answer built only from facts already on the page.	Yes, within the joint section
3. Structured facts	Restates existing facts as labelled fields (departure, destination, starting fare and fees), keeping qualifying words such as “from”.	Yes, within the joint section
4. Schema with visible structure	Adds missing schema.org fields to the page’s existing machine-readable data, together with a matching visible layout.	No
5. External facts with attribution	Adds a statistic from an outside source with a clear credit, kept separate from any fare claims.	No
6. Automatic AI rewriting	Asks the AI to rewrite the page and rejects any rewrite that adds a new number. This is our own rewriting method, not a replication of the published AutoGEO system.	No
Position test	Moves Flight Centre’s unchanged page to the top of the AI’s list of sources.	No: a website owner cannot control this

7.2 THE FINAL PAGE, AND WHAT FLIGHT CENTRE WOULD DO WITH IT

The final implementation has four technical operations, grouped into three content changes in the leadership deck: rendered deals, a joint question-and-facts block, and restored/repositioned FAQs. These counts describe the same stack. The sandbox changes saved representations; production changes should also remain useful and accurate for customers.

TABLE 8 The four changes in the final page

CHANGE	WHAT IT DOES	HOW FLIGHT CENTRE CAN CHECK IT	WHO ACTS
Rendered content and organisation (all 52 pages)	Uses the page as it looks after JavaScript runs, tidies the sections and fixes broken price spacing. A check confirmed that no other text changed.	Load the page with JavaScript turned off and see whether the deal text is there.	Front-end developers

CHANGE	WHAT IT DOES	HOW FLIGHT CENTRE CAN CHECK IT	WHO ACTS
Joint FAQ-and-facts section (27 of 52 pages)	Adds a route-specific question and labelled facts on 27 of 52 pages. The other 25 did not receive this joint block; separate rendered-content or FAQ transformations may still apply.	Check that every field matches the live page and current pricing.	Content team, with pricing and legal review
Restored FAQ questions (429 questions)	FAQ questions sit inside accordion buttons, which text extraction strips out. This change turns them into plain text next to their answers, keeping links and conditions.	View the page source and check that each FAQ question appears as text.	Front-end developers
FAQ placement (9 pages)	Moves the FAQ section above the general "Explore" destination section.	Check the order of sections in the page source.	Front-end developers

The joint FAQ-and-facts section always follows the same simple template:

```
<section id="geo-final-route">
  <h2>What starting fare and fees does this page state for [origin] to [destination]?</h2>
  <dl>
    <dt>Departure</dt><dd>[origin] ([airport code])</dd>
    <dt>Destination</dt><dd>[destination] ([airport code])</dd>
    <dt>Starting fare stated on page</dt><dd>from [fare]</dd>
    <dt>Fees</dt><dd>[the page's own fee notice, original markup kept]</dd>
  </dl>
</section>
```

The first two changes were sealed as release r4 on 20 September 2026 (`final_release.py`), and the two FAQ fixes were added as release r5 (`flight_first_refinement.py`). The final page deliberately leaves out added schema, external facts, position changes, AI rewriting and the other experimental options.

08 RESULTS

The final page raised citation inclusion from 40.0% to 49.3% in its first run and to 52.4% in its repeat run, and both gains have 95% intervals above zero.

8.1 INDIVIDUAL METHODS

Figure 2 shows the effect of each method when it was tested on its own. Structured facts, the grounded FAQ and schema with visible structure all stayed clear of zero. Rendered content and external facts were positive but uncertain, and AI rewriting clearly made things worse. These gains cannot simply be added together. For external facts, the small overall gain is misleading: in the 80 prompts where the added facts actually changed what the AI read, there was no net gain.

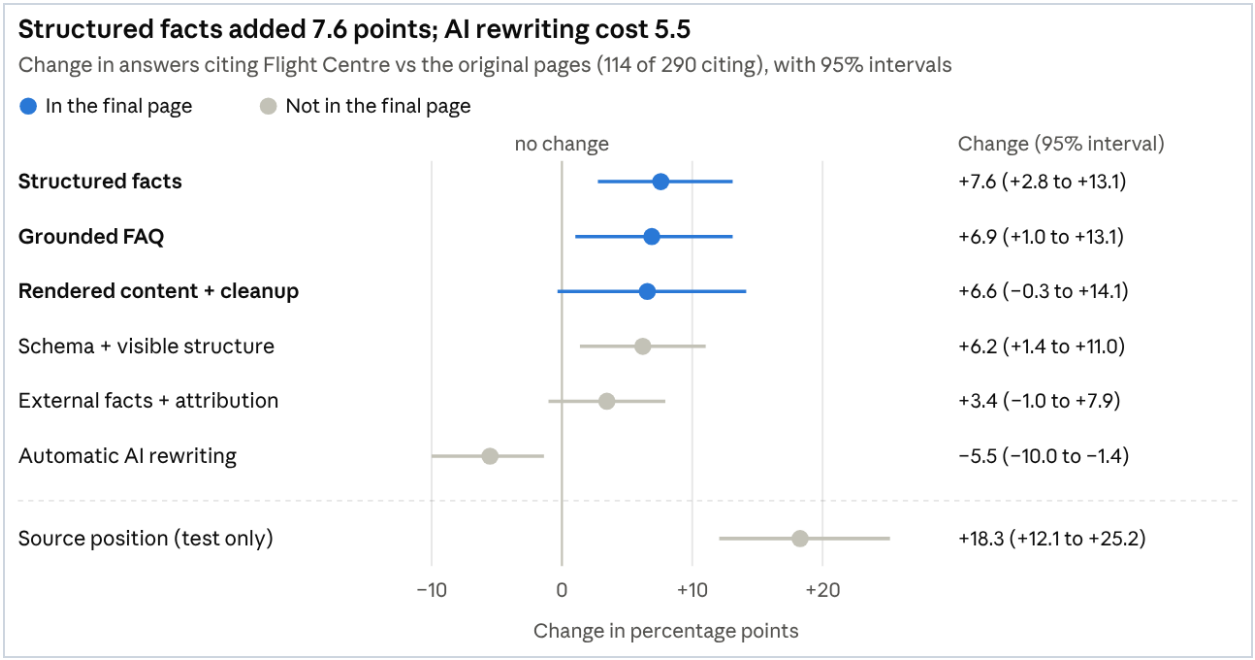


FIGURE 2 Change in the share of answers citing Flight Centre for each method tested on its own, compared with the original pages (114 of 290 answers citing). Lines show 95% intervals.

METHOD BY METHOD

Each card below shows one method tested on its own against the original pages, with its 95% interval. The label shows whether the whole interval sits above zero, crosses zero or sits below zero. These intervals are exploratory (Section 6.5), so they show the direction and rough size of each effect rather than a confirmed result.

+7.6
PERCENTAGE POINTS
95% interval +2.8 to +13.1
INTERVAL ABOVE ZERO

STRUCTURED FACTS

What changed. Facts already on the page (departure, destination, starting fare and fees) were restated as labelled fields, keeping qualifying words such as "from".

What it means for Flight Centre. This was the strongest single change we tested. A short block of labelled facts on route pages is worth piloting, as long as every field is checked against live pricing.

In the final page: Yes, inside the joint FAQ-and-facts section

+6.9
PERCENTAGE POINTS
95% interval +1.0 to +13.1
INTERVAL ABOVE ZERO

GROUNDED FAQ

What changed. One question and answer, built only from facts already on the page.

What it means for Flight Centre. Presenting page content as the question a customer would ask helped. The answer must come from the page itself, because AI-written answers can invent facts.

In the final page: Yes, inside the joint FAQ-and-facts section

+6.6
PERCENTAGE POINTS
95% interval −0.3 to +14.1
INTERVAL CROSSES ZERO

RENDERED CONTENT AND CLEANUP

What changed. The version of the page captured after JavaScript runs, so deal cards and fares became readable text, with tidied sections and repaired prices.

Its standalone interval crosses zero. Making useful facts available without JavaScript can improve accessibility, but crawler capabilities vary. The historical Vercel study does not establish how every current engine fetches a page.

In the final page: Yes, as the base of the final page

+6.2
PERCENTAGE POINTS
95% interval +1.4 to +11.0
INTERVAL ABOVE ZERO

SCHEMA WITH VISIBLE STRUCTURE

What changed. Missing schema.org fields were added to the page's existing machine-readable data, together with a matching visible layout.

What it means for Flight Centre. The interval sat above zero, but we could not separate the effect of the schema from the visible layout that came with it. Google says its AI features need no special markup (Google Search Central, 2025), so we do not recommend adding schema for its own sake.

In the final page: No

+3.4
PERCENTAGE POINTS
95% interval −1.0 to +7.9
INTERVAL CROSSES ZERO

EXTERNAL FACTS WITH ATTRIBUTION

What changed. A statistic from an outside source, with a clear credit, kept separate from any fare claims.

What it means for Flight Centre. The small overall gain is misleading. In the 80 prompts where the added facts actually changed what the AI read, there was no net gain, so we do not recommend this method on current evidence.

In the final page: No

−5.5
PERCENTAGE POINTS
95% interval −10.0 to −1.4
INTERVAL BELOW ZERO

AUTOMATIC AI REWRITING

What changed. The AI rewrote the page, and any rewrite that added a new number was rejected.

What it means for Flight Centre. Even with that safeguard, rewritten pages were cited less often. Flight Centre should not rewrite pages automatically for AI engines. Changes should be written and reviewed by people.

In the final page: No

+18.3
PERCENTAGE POINTS
95% interval +12.1 to +25.2
BENCHMARK ONLY

SOURCE POSITION (TEST ONLY)

What changed. Flight Centre's unchanged page was moved to the top of the AI's list of sources.

What it means for Flight Centre. Position was the biggest lever we measured, which is why every comparison keeps the source order fixed. A website cannot choose where an AI engine places it, so this is a benchmark, not a recommendation.

In the final page: No: a website owner cannot control it

8.2 THE FINAL PAGE

Table 9 compares the final page with the original pages in both runs.

TABLE 9 Final page results (290 answers per version)

RUN	ANSWERS CITING FLIGHT CENTRE	RATE	CHANGE FROM ORIGINAL	95% INTERVAL	QUESTIONS UP / DOWN / SAME
Original pages	116 of 290	40.0%	–	–	–
Final page, first run	143 of 290	49.3%	**+9.3 points**	+3.1 to +15.9	15 / 4 / 10
Final page, repeat run	152 of 290	52.4%	**+12.4 points**	+5.2 to +20.0	21 / 6 / 2

The two runs used identical pages, instructions and sources, so the 3.1-point difference between them (interval −2.8 to +8.3) is simply the AI answering differently each time. The repeat run is not "better"; it is the same result measured twice. In relative terms, 23% and 31% more answers cited Flight Centre. The final rate is 52.4%, which is not the same as a 52.4% improvement.

8.3 CHECKS OUTSIDE THE MAIN TEST

We ran two further checks to see whether the gain holds outside the main test, as well as checks that the test itself worked as intended (Table 10). Neither of the two further checks was conclusive.

TABLE 10 **Checks outside the main test**

CHECK	RESULT	WHAT IT TELLS US
Secondary setup, where the system finds passages itself	26 of 400 answers before, 27 of 400 after (6.50% to 6.75%)	There was no clear gain when Flight Centre first had to be found, so the study does not show better discovery.
29 reworded questions (9 October), with original and final pages tested side by side	199 to 223 of 290 (68.6% to 76.9%): +8.3 points, interval −1.0 to +17.9, p = 0.12	The estimate is positive but its interval crosses zero. This check does not independently confirm an improvement.
Implementation checks	Source order and competitor text were identical in every pair; all 330 Flight Centre texts sent to the AI match the saved pages; the final FAQ stage rebuilds all 52 pages exactly.	The AI read exactly the pages we meant to test, and the last step can be reproduced.

8.4 WHAT THE PERCENTAGES MEAN FOR FLIGHT CENTRE

In the selected comparison, FC received at least one valid citation in 116 of 290 baseline answers and 152 of 290 optimised answers. That is 36 more citing answers, or +12.4 percentage points. It is not a count of explicit brand recommendations, customers, visits or bookings.

The headline population is 29 eligible questions. That does not mean FC was absent for all other 21: the saved search audit reports top-10 coverage for 30 of 50 questions, including one outside the main test. The saved-source result does not establish better live retrieval or commercial performance.

09 EVIDENCE FROM THE PROJECT

Figures 3 to 10 follow one real question through the sandbox: “I’m looking for flights from Melbourne to London” (question Q06). The code shown is real and comes from the repository, with “…” marking lines we skipped.

queries/queries.csv

fx i’m looking for flights from melbourne to london

	A	C	E
1	**id**	**group**	**query**
2	Q01	Flights	i’m looking for flights to melbourne
3	Q02	Flights	i’m looking for flights from sydney to melbourne
4	Q03	Flights	i’m looking for flights from sydney to london
5	Q04	Flights	i’m looking for flights from australia to london
6	Q05	Flights	i’m looking for multi city flight tickets
7	**Q06**	**Flights**	**i’m looking for flights from melbourne to london**
8	Q07	Flights	i’m looking for flights from perth to bali
9	Q08	Flights	i’m looking for flights from australia to the united states

FIGURE 3 The question file. Each row is one question, written the way a customer would ask it. Question Q06 is highlighted.

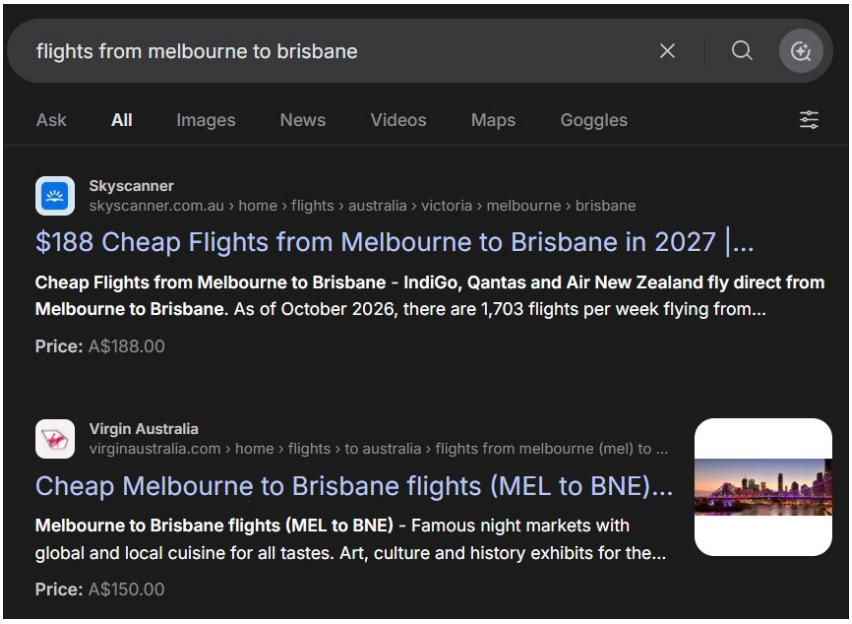


FIGURE 4 Brave Search results for a travel question. Our code saved results like these for all 50 questions in August 2026.

geo_bench/retrieval/cleaner.py

```
BOILERPLATE_TAGS = ["script", "style", "noscript",
    "nav", "header", "footer", "aside", "form",
    "button", "svg", "iframe"]
...
def clean_html(html: str) -> Tuple[str, str]:
    soup = BeautifulSoup(html, "html.parser")
    jsonld = extract_jsonld(soup)
    for tag in soup(BOILERPLATE_TAGS):
        tag.decompose()
...
    text = re.sub(r"\s+", " ",
        soup.get_text(" ", strip=True)).strip()
    return text, jsonld
```

SAVED TEXT FOR MELBOURNE → LONDON (AUG 2026)

```
Cheap Flights from Melbourne (MEL) to London (LON) from $1,434
Flights from Melbourne to London $1,434 ^ Hot deals for a London
getaway! Displayed fares exclude Online Booking Fee & Merchant Fee
. Fees are applied once at checkout. …
```

FIGURE 5 The cleaning code removes scripts, menus and buttons and keeps the text the AI will read (top), shown with the saved text of the Melbourne–London page (bottom). Because buttons are removed, FAQ questions inside accordions disappear at this step, which is the problem the FAQ fix solves.

optimisation_v3/structured_refinement.py

```
# Unknown qualifications are never silently discarded.
if not fee or fee.end() != len(e.quote):
    ...
    continue
...
fields = [
  ("Departure", m["origin"]+" ("+m["origin_code"]+")"),
  ("Destination",
     m["destination"]+" ("+m["destination_code"]+")"),
  ("Starting fare stated on page", "from "+m["fare"]),
  ("Fees", fee.group())]
```

ADDED TO THE FLIGHT CENTRE PAGE

```
Departure: Melbourne (MEL)
Destination: London (LON)
Starting fare stated on page: from $1,434
Fees: Displayed fares exclude Online Booking Fee …
```

FIGURE 6 The structured-facts code copies facts that are already on the page into labelled fields and skips any layout it does not recognise (top). The fields it added to the Melbourne–London page are shown at the bottom.

BASELINE • PAGE OPENING

Flights from Melbourne to London

$1,434 ^

Displayed fares exclude Online Booking Fee & Merchant Fee.

Fees are applied once at checkout.

Baseline headline; not an individual offer.

OPTIMISED • OFFER RECORD

Melbourne (MEL) to London (LON)

AIRLINE
China Eastern

FARE AND TRIP TYPE
$1387* Economy, Return

AVAILABLE BETWEEN
08 Feb 27 – 11 Mar 27

Later rendered offer; not a like-for-like fare comparison.

FIGURE 7 Offer information before and after: the original page opening (left) and the rendered offer record used in the improved page (right). The two were captured on different dates, so this is not a price comparison.

BASELINE • MODEL INPUT

Frequently asked questions

We make travel easy with excellent service, irresistible deals & the widest choice of travel options for every traveller. We give you peace of mind when you need it most and have expertise that you only get from 40 years in the business!

OPTIMISED • MODEL INPUT

Frequently asked questions

Why should I book a flight through Flight Centre?

We make travel easy with excellent service, irresistible deals & the widest choice of travel options for every traveller. We give you peace of mind when you need it most and have expertise that you only get from 40 years in the business!

FIGURE 8 The FAQ text the AI received. In the original page (left) the question is missing because it sat inside a button. In the improved page (right) it is restored next to its answer.

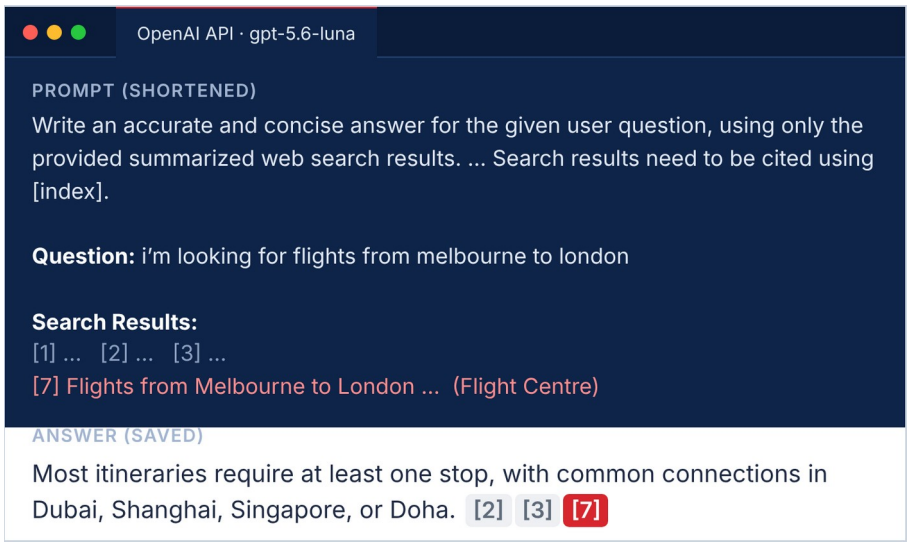


FIGURE 9 The instruction sent to the AI with its numbered sources (top), and the saved answer, in which [7] is the Flight Centre page (bottom).

03_code/verify_results.py

```python
def inclusion(r):
    ids = {int(x) for x in
           re.findall(r"\[(\d+)\]", r["answer"])}
    return int(any(0 < i <= len(r["sources"])
        and r["sources"][i-1]["brand"] == "flightcentre"
        for i in ids))
```

TERMINAL

```
$ python scripts/verify_public_counts.py
initial_minus_baseline_95_pp [3.103448275862069, 15.862068965517242]
repeat_minus_baseline_95_pp [5.172413793103448, 20.0]
repeat_minus_initial_95_pp [-2.7586206896551726, 8.275862068965518]
PASS: question-level totals 116 / 143 / 152 out of 290.
```

FIGURE 10 The scoring function (top) and the real output of the public check script, which reproduces the totals of 116, 143 and 152 (bottom).

A complete saved answer. The answer below was saved for question Q06 with the improved page (source order 2, repetition 4) and is shown unedited. Reference [7] is Flight Centre's Melbourne–London page (flightcentre.com.au/flights/au-vic-melbourne/gb-eng-london).

> **SAVED ANSWER · Q06 · IMPROVED PAGE · [7] = FLIGHT CENTRE**
>
> I can help you compare Melbourne-London flights, but I'll need your travel dates, whether you want one-way or return, number of passengers, and preferred cabin class.
>
> Indicative fares currently include one-way options from about A$648 and return fares from approximately A$1,096-A$1,386, although prices and availability can change quickly. [1]**[7]**[8]
>
> Most itineraries require at least one stop, with common connections in Dubai, Shanghai, Singapore, or Doha. [2] [3]**[7]**
>
> Airlines serving the route include Emirates, Etihad Airways, Qatar Airways, China Eastern, Singapore Airlines, Qantas, British Airways, and Finnair. [1][2][3]
>
> The main departure airport is Melbourne Airport (MEL), while London options include Heathrow (LHR), Gatwick (LGW), Stansted (STN), Luton (LTN), and London City (LCY). [1][4]**[7]**

For the same point about stopovers, the matching answer for the original page cited [2][3][5][8] and no Flight Centre source. Every saved answer, with its full numbered list of sources and their page text, is in `04_evidence/selected` in the ZIP and can be browsed in the results explorer.

PART

04

HOW FAR THE RESULTS GO

How we tested the work, the problems we solved, the limits of the evidence, and what we learned.

> **The findings support content improvements. They are not a forecast of live traffic or bookings.**

IN THIS PART

10	Testing and validation	25
11	Technical challenges	25
12	Limitations	26
13	Project history and lessons learned	26

10 TESTING AND VALIDATION

This handover reran offline record/count checks. Historical unit-test results describe their original runs; they are not a new complete suite certification. See Test_Plan_and_Validation_Record.md for current checks and unresolved human review.

TABLE 11 Testing at each level

LEVEL	WHAT IT CHECKS	RESULT
Unit tests for the final page code (12 test files)	That the code never invents facts, leaves unfamiliar pages alone, rejects tampered inputs, and restores and moves FAQs correctly	48 of 55 passed and none failed; 7 need the ZIP's evidence files (re-run on 21 October 2026)
Unit tests for the seven method snapshots	Each method's own safety checks	171 of 208 test runs passed and none failed; the other 37 need the same files
Benchmark checks (14 tests)	Scoring, statistics, hiding brand names from the quality judge, source shuffling, prompt limits and fingerprints	All 14 passed on both versions of the benchmark code
Fairness checks	The four controls (T1–T4) and the pair check on all 290 final pairs	All passed
Offline audits	A recount of every saved answer, a check of the text sent to the AI, and a rebuild of the final FAQ stage	All passed, reproducing 116, 143 and 152

Changes to the settings, the data and the experiment plan are recorded with their dates: in the decisions log for the baseline phase (August 2026), and in dated protocol files, such as the r5 protocol of 20 September 2026, for the later runs. Two checks are still outstanding: a human review of whether each citation genuinely supports its sentence, and a calibration of the AI quality judge against two human graders (target agreement $\kappa \geq 0.6$).

11 TECHNICAL CHALLENGES

Most of our challenges came from turning a constantly changing, partly hidden system into something that could be tested fairly. Table 12 lists the main ones and how we solved them. Each fix is recorded with its date in the decisions log and the protocol files.

TABLE 12 Main technical challenges and how we solved them

CHALLENGE	HOW WE SOLVED IT
Live AI engines cannot be tested fairly: pages cannot be swapped inside them, their answers vary, and we cannot see which pages they read.	We built the frozen sandbox, where only Flight Centre's page changes (August 2026).
Bing's search API was retired in August 2025, so it could not be our search engine.	We used the Brave Search API instead, and used Bing once as a cross-check: Flight Centre appeared in both engines' top 10 for 21 of the 50 questions.
Deal and fare content needs JavaScript, and some competitors block bots. Expedia blocked AI crawler identities on all 31 of its pages.	We captured the original pages without JavaScript, used one browser identity for every brand, and fell back to a Common Crawl copy when a page was blocked. The rendered Flight Centre content became one of the methods.
A recording bug: for 173 of the first 180 answers, the source numbers were saved in the wrong order.	We fixed the recording, discarded every earlier answer, generated them again from the same snapshot and tightened the protocol lock.

CHALLENGE	HOW WE SOLVED IT
Too few questions for reliable statistics, and an early design that sent only one passage per page, which hid content added elsewhere on the page.	We expanded the snapshot to 50 questions, switched the main setup to whole pages of up to 24,000 characters, and generated the original-page answers again.
FAQ questions were stripped out during cleaning, and some prices were split up by stray spaces.	We converted expanded, question-shaped accordion titles into plain text, and limited the price fix to complete price elements, with a check that no other text changed.
API credits ran out partway through a 580-answer run, and testing many variants risked over-fitting.	The scripts resume only the missing answers under the same settings. We kept every variant as a separate copy of the code and reported the failed methods and both final runs.

12 LIMITATIONS

Our results apply to this sandbox, these questions and this AI model. They show that the way a page presents information affects whether it is cited. They do not show that live visibility or sales will rise. Table 13 sets out each limitation and what it means for our conclusions.

TABLE 13 Limitations and what they mean for our conclusions

LIMITATION	WHAT IT MEANS
One search snapshot, one saved model configuration and 29 questions with FC already in the supplied source pool.	We have no evidence about content freshness, other markets, other engines, or questions where Flight Centre does not appear. The secondary setup found no gain in discovery.
The original-page answers were generated earlier than the final-page answers, and the final page was chosen after testing many variants.	Changes in the AI model over time cannot be ruled out, and the headline result is probably somewhat optimistic. The side-by-side check with reworded questions found a smaller, unconfirmed gain (+8.3 points).
Only 29 questions and related intents limit precision and generalisation.	Smaller real effects may have been missed.
We measured whether Flight Centre was cited, not whether the citation was accurate, a recommendation or clicked, and the AI quality judge was not calibrated.	Commercial value was not measured, and a human review of the citations is still outstanding.
The rendered pages were captured on different dates, so offers are not price-matched. The joint section fitted only 27 of the 52 pages, and the gain cannot be split between the four changes.	Part of the gain may come from different content rather than presentation alone. Other page templates need new rules, and we cannot say which change did the most.
We have no data on live rankings, traffic, bookings or revenue.	Monitor commercial value separately.

13 PROJECT HISTORY AND LESSONS LEARNED

The project ran over two semesters in 2026. In Semester 1 (IFN735) we researched the problem and audited Flight Centre's website. In Semester 2 (IFN736) we built the sandbox, tested the methods and prepared this handover. Table 14 lists the main milestones.

TABLE 14 Project milestones

WHEN	MILESTONE
Semester 1	Literature review; audits of Flight Centre's website (crawlability, structured data and FAQ accordions); a live check of 89 everyday questions, which found no Flight Centre citations; interim report delivered.
27 July – 3 August	Sprint 2: the sandbox worked end to end for the first time.
10 – 16 August	First Brave snapshot captured and the test plan locked; data-integrity audit; snapshot expanded to 50 questions; whole-page baseline signed off.
Late August – September	Six methods built and tested one at a time, each with its own owner, unit tests and frozen copy of the code.
20 September	Individual and combined results audited; the final page sealed and its test plan frozen before the final run.
21 – 29 September	Final page tested, then tested again with identical inputs.
9 October	Check with 29 reworded questions.
18 – 19 October	Independent recount of both final runs; claim–evidence ledger and public repository published.
Week 13	Final presentation, artefact submission and handover to Flight Centre.

13.1 WHAT CHANGED BETWEEN THE SEMESTERS

Semester 1 planned a broader evaluation than we could complete, including AI-judged answer quality and outside validation. We built those measures but could not calibrate them in time, so we do not rely on them. We also corrected two Semester 1 statements after checking official documentation: FAQ schema is not required for Google's AI features, and not every AI crawler ignores JavaScript. Finally, the Semester 1 result of no citations from 89 questions came from live engines without supplied sources, so it cannot be compared with the sandbox rates.

13.2 LESSONS LEARNED

1. **Freeze before you test.** Nothing could be compared until the snapshot was frozen, which is why we abandoned our earlier private collection of pages.
2. **Audit before you report.** Our August audit caught the source-number bug before it could affect any result we reported.
3. **Give the AI whole pages, not snippets.** A one-passage design could never see FAQs or facts added elsewhere on the page.
4. **Decide the main measure in advance, and report what happened to it.** Our pre-declared primary measure did not move. Saying so openly is what makes the positive result credible.
5. **Budget for repeats.** Answer-to-answer variation was the biggest source of noise, and re-running the final page was the most informative extra test we did.
6. **Explain what a number means.** "52.4%" without "not a 52% improvement" can easily mislead a business reader.

PART

05

USING AND HANDING OVER THE WORK

How Flight Centre can use, check and extend the sandbox, the obligations that come with it, and how the handover works.

> **Three of the four ways to use the artefact need no API keys and cost nothing.**

IN THIS PART

14	User guide	29
15	Maintaining and extending the sandbox	31
16	Ethical, legal and professional considerations	32
17	Handover, intellectual property and licensing	33
18	Use of generative AI	33

14 USER GUIDE

Flight Centre can use the artefact at four levels (Table 15). The first three need no API keys and cost nothing.

TABLE 15 **Four ways to use the artefact**

LEVEL	WHAT YOU CAN DO	WHAT YOU NEED
1. Explore	Browse every question's answers and sources, and compare original and improved pages	A web browser
2. Verify	Recount the headline results from the saved answers	Python 3.10 or later
3. Rebuild	Rebuild the final FAQ stage for all 52 pages and check the fingerprints	Python, plus the beautifulsoup4 and requests packages
4. Run a new test	Capture a new snapshot of current pages and compare page versions	A Python developer, your own OpenAI and Brave Search API keys, and an API budget

14.1 LEVEL 1: EXPLORE THE RESULTS

1. Unzip the assessment package and open START_HERE.html in a web browser.
2. Open the results explorer (`04_evidence/Results_Explorer.html`) to read every question's answers with their numbered sources. It includes questions where citations went down, not only those where they went up.
3. Compare an original and an improved page in the `04_evidence/pages` folder.

14.2 LEVEL 2: VERIFY THE NUMBERS

From the top level of the public repository, run the quick check:

```
python scripts/verify_public_counts.py
```

It should finish with `PASS: question-level totals 116 / 143 / 152 out of 290`. For a stronger check that rescores every saved answer, run these two scripts from the top level of the ZIP:

```
python 03_code/verify_results.py
python 03_code/verify_repeated_runs.py
```

The first prints a summary showing `"status": "PASS"` after checking all 290 answer pairs, the seven individual results, the fingerprints of the 52 pages and the 330 Flight Centre texts sent to the AI. The second ends with `PASS: 116 baseline, 143 initial, 152 repeat`.

14.3 LEVEL 3: REBUILD THE FINAL FAQ STAGE

```
pip install -r 03_code/refinement_requirements.txt
python 03_code/rebuild_final_refinement.py
```

The script should report PASS for all 52 pages and save its result to `02_documentation/Refinement_Rebuild.json`.

14.4 LEVEL 4: RUN A NEW TEST

This level uses paid APIs. Each AI answer is one paid call, a single before-and-after comparison needs 580 answers, and capturing a new snapshot takes about 150 searches.

1. **Set up Python.** In `03_code/selected_snapshot`, create a virtual environment and install the packages listed below.

```
python -m venv .venv
source .venv/bin/activate          # on Windows: .venv\Scripts\activate
pip install requests beautifulsoup4 numpy scipy rank-bm25 openai
```

2. **Add your own API keys.** Create a file called `.env` in the same folder containing the two lines below. The settings file reads it automatically. Never commit this file to the repository, and set a spending limit with each provider. To use Claude models instead, add `ANTHROPIC_API_KEY`, install the anthropic package and set the mode to "anthropic".

```
OPENAI_API_KEY=your-openai-key
BRAVE_API_KEY=your-brave-key
```

3. **Add your questions** to `queries/queries.csv`, using the columns id, stratum, group, intent, query, source and corpus_note.
4. **Check the settings** in `geo_bench/config.py` (mode, model and region), and record any change in the decisions log. For a free dry run, set the mode to "offline" and add `--allow-demo`; demo results are never counted.
5. **Capture and lock a new snapshot.** Stop if any of the control checks fail.

```
python -m geo_bench freeze-live
python -m geo_bench verify   --from-run <RUN_ID>
python -m geo_bench qc       --from-run <RUN_ID>
python -m geo_bench lock     --from-run <RUN_ID>
python -m geo_bench route    --from-run <RUN_ID>
python -m geo_bench index baseline --from-run <RUN_ID>
python -m geo_bench controls --from-run <RUN_ID>
```

6. **Generate and score the original-page answers.**

```
python -m geo_bench run baseline --variant fixed --from-run <RUN_ID>
python -m geo_bench score baseline --variant fixed
python -m geo_bench stats --variant fixed
```

7. **Apply the page changes** in `optimisation_v3` (first `final_release.final_core`, then `flight_first_refinement.refine`) to the new Flight Centre pages, and answer the same 290-answer schedule. The original run scripts (`optimisation_v3/experiment.py` and `refinement_experiment.py`) show how the matched comparison was built, but they point to the original result folders, so a developer will need to point them at the new run.
8. **Compare fairly.** Generate the original and improved answers in the same period, count citation inclusion for each question, use the paired question bootstrap, and report every run.

Results are saved in `runs/<RUN_ID>` (the snapshot and its lock file) and in the `results` folder (answers, scores and quality checks).

14.5 TROUBLESHOOTING

- If a run stops because of configuration or endpoint drift, a setting (or `OPENAI_BASE_URL`) changed after the snapshot was locked. Lock the snapshot again and record the change.

- If you see “Use a new release path; no overwrite”, the output folder already exists. Choose a new folder. Research reuse is optional.
- If `verify_results.py` cannot find `04_evidence`, you are running it from the public repository. Run it from the top level of the ZIP instead.

15 MAINTAINING AND EXTENDING THE SANDBOX

The sandbox was built to be reused. This section explains how to keep it current, how to add a new method and what we recommend Flight Centre does next.

15.1 KEEPING THE SANDBOX UP TO DATE

- Capture a new snapshot and run a new baseline whenever Flight Centre’s page templates change, or before any new study. Never mix results from different snapshots.
- When the AI model changes, re-run both the original and the improved pages under the new model.
- Record every change in the decisions log, and re-run the unit tests after any code change with `python -m unittest discover -s optimisation_v3 -p "test_*.py" -t .` from `03_code/selected_snapshot`.
- Keep API keys only in the `.env` file or a secrets manager.

15.2 ADDING A NEW METHOD

1. Write it as a function in `optimisation_v3` that returns the new page with its fingerprints, following the `finish()` function in `final_release.py`.
2. Build in the same safeguards: reuse only facts already on the page, leave unfamiliar layouts unchanged, and never read the test questions.
3. Add unit tests beside it, including one showing that an unfamiliar page is left unchanged and one showing that no new facts are added.
4. Save it as a separate copy of the code and test it against the same original pages.

15.3 RECOMMENDED NEXT STEPS

Table 16 lists our recommended next steps, in the order we suggest taking them.

TABLE 16 Recommended next steps

STEP	WHAT TO DO	HOW TO JUDGE SUCCESS
1	Serve deal, fare and FAQ text directly in the page HTML, using server-side rendering or pre-rendering, with FAQ questions as plain text.	Published template text contains the approved facts without requiring JavaScript.
2	Apply the labelled question-and-facts section where it fits the route template.	Every field matches the live page and passes pricing and legal review.
3	Monitor available platform reports, dated answer examples, referred visits, enquiries and quotes. Verify real crawler access in logs.	Reporting distinguishes citations, brand mentions and commercial outcomes; sampled citations are checked for support.
4	Prioritise ongoing template improvements and assess vendors or app/API integrations on their business case. Repeating the sandbox is optional research.	Owners and outcomes recorded.
5	Keep research limitations explicit. Human citation review and judge calibration remain outstanding if quality scores will be used.	Document reviewers, disagreements and calibration evidence before relying on quality scores.

The sampled rendering evidence supports making useful commercial facts accessible without JavaScript. Platform capabilities vary. Appropriate structured data should match visible content; no special AI schema guarantees citation. See Appendix C for FC implementation priorities and source references.

16 ETHICAL, LEGAL AND PROFESSIONAL CONSIDERATIONS

The study used public pages and generic travel queries. Captures can contain public names or other third-party information, so the full archive remains a private handover. Historical prices are not current offers. Live deployment needs normal privacy, commercial, legal and content review.

TABLE 17 How the project meets ethical, legal and professional obligations

FRAMEWORK	HOW THE PROJECT MET IT	WHAT FLIGHT CENTRE SHOULD KEEP DOING
Australia's AI Ethics Principles (Department of Industry, Science and Resources, n.d.) and the **Guidance for AI Adoption** (National AI Centre, 2025)	We reported every result with its uncertainty, including the methods that failed. Methods were tested before use, the decisions log records who changed what and why, and the code lists every added claim for human sign-off.	Give someone clear ownership of GEO work, monitor how AI engines cite Flight Centre, and approve changes before publishing.
Australian Consumer Law on false or misleading claims (ACCC, n.d.)	The selected stack uses saved FC facts and fee notices. External-facts arms are separate. Different captures contain different fares; current offers require fresh validation.	Check every fare, fee and condition against live pricing before publishing.
Privacy Act 1988	We used generic travel questions and public pages, and sent no customer or confidential data to any AI service.	Do not send customer or confidential data to AI tools without approval under Flight Centre's AI policy.

FRAMEWORK	HOW THE PROJECT MET IT	WHAT FLIGHT CENTRE SHOULD KEEP DOING
Google's spam policies (Google Search Central, n.d.)	All final-page changes are visible to customers. Keyword stuffing and padding were only used as research controls and have been retired.	Never show AI crawlers different content from what customers see.
Copyright, website terms and the ACS Code of Professional Ethics (Australian Computer Society, 2023)	Third-party pages were captured for research only and are not republished. Limitations are stated plainly, and published research is credited.	Keep the snapshot internal, and check the Brave Search API terms and website terms before any reuse.

17 HANDOVER, INTELLECTUAL PROPERTY AND LICENSING

Flight Centre will receive everything it needs to continue the work without the team. No passwords or API keys change hands. Table 18 sets out the handover arrangements.

TABLE 18 Handover arrangements

ITEM	ARRANGEMENT
What is delivered	The public GitHub repository (github.com/muhammadmubbashar/GEO--Generative-Engine-Optimisation) and the assessment ZIP (`GEO_Semester_2_Assessment_Package.zip`) containing the full evidence.
When and how	The team will hand both over to Flight Centre's project contact at the final project meeting, before the Week 13 submission deadline, with a walkthrough of START_HERE.html, the results explorer, the check scripts and this guide.
Credentials	None are handed over. Flight Centre uses its own OpenAI and Brave Search API keys (Section 14.4).
Support	No support is planned after handover. This guide, the Test Plan and the project records are written so that Flight Centre can continue on its own.
Acceptance	We will ask Flight Centre to confirm receipt and acceptance by email.

Intellectual property and licensing. The code is released under the MIT licence (copyright 2026 Muhammad Mubbashar). As set out in the Artefact Agreement, ownership and licensing will be confirmed with Flight Centre and QUT before handover, in line with QUT's intellectual property policy. The improved pages are derived entirely from Flight Centre's own content. Captured competitor and third-party pages remain the property of their owners and are included only so that the results can be verified. The AI answers were generated under OpenAI's terms of use, and the instruction given to the AI comes from the published GEO paper (Aggarwal et al., 2024).

18 USE OF GENERATIVE AI

Generative AI was both the subject of this experiment and a tool we used to prepare it. The team reviewed and checked all AI-assisted work and takes responsibility for it. No Flight Centre confidential or customer data was entered into any AI tool. Table 19 lists each way we used it.

TABLE 19 How generative AI was used

HOW AI WAS USED	TOOL	NOTES
As the system being tested: answering travel questions, writing search variations, the AI rewriting method and the answer-quality judge	OpenAI API (model gpt-5.6-luna)	Used with the provider's default settings. The judge was not calibrated and is not used for conclusions.
Meaning-based search in the secondary setup	OpenAI text-embedding-3-small	Secondary setup only.
Writing the 29 reworded questions for the 9 October check	An AI assistant	Listed as a limitation, because these are not real customer wordings.
Drafting this report and the Test Plan; preparing Figures 1–3, 5, 6, 9 and 10 from the repository's code, data and saved outputs; calculating the detection thresholds in Section 6.2	Claude (Anthropic, 2026)	Every number was checked against the repository files and re-run scripts, and the text was reviewed by the team.
Preparing presentation slides and speaker scripts	Claude (Anthropic, 2026)	Checked against the results files.
Helping with code scaffolding and repository documentation	AI coding assistants	Checked by the unit tests, fairness checks and offline audits.

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APPENDIX A. GLOSSARY

TABLE 20 Glossary of terms

TERM	MEANING
Citation inclusion	The share of answers that cite Flight Centre at least once. Several citations in one answer count once.
Percentage point	The plain difference between two rates. Going from 40.0% to 52.4% is a rise of 12.4 percentage points.
Relative increase	The change compared with the starting value. 36 more citing answers on top of 116 is 31% more.
95% interval	The range of gains that fits the data, found by resampling whole questions. If it crosses zero, the gain is uncertain.
Original pages (baseline)	Flight Centre's pages as captured, before any change.
Snapshot	The dated, fingerprinted copy of search results and pages that every test runs on.
Main (fixed) setup	The setup behind the headline results, where the same whole pages are given to the AI in both versions.
Secondary (reindexed) setup	A setup where the system searches all captured pages for relevant passages itself.
GEO	Generative engine optimisation: improving content so that AI answer engines use and cite it.
No-op	When a method leaves a page completely unchanged because it does not recognise the layout.
Fingerprint (SHA-256)	A short code calculated from a file's contents. It changes if even one character changes.
PAWC	Position-adjusted word count, the GEO paper's visibility score. It is a secondary measure in this study.

APPENDIX B. WHERE TO FIND THINGS

TABLE 21 Where to find the code and records

WHAT	FILE OR FOLDER	LOCATION
Questions: 04_evidence/queries/original_queries.csv; saved query subsets alongside it.		
Page capture, cleaning and quality checks	geo_bench/retrieval/cleaner.py; geo_bench/crawler/quality.py; geo_bench/experiment/run_store.py	03_code/selected_snapshot
Page changes and their unit tests	optimisation_v3 (final_release.py, flight_first_refinement.py, rendered_organisation.py, structured_refinement.py, faq_refinement.py, arms.py and the test_ files)	03_code/selected_snapshot
AI answering, scoring and statistics	geo_bench/generation; geo_bench/api; geo_bench/evaluation; geo_bench/baseline_v2.py	03_code/selected_snapshot
All experimental settings and the final run plan	geo_bench/config.py; FLIGHT_FIRST_R5_PROTOCOL.md	03_code/selected_snapshot
One frozen copy of the code per method	individual_snapshots (one folder per method, named after it)	03_code
Offline checks and the FAQ rebuild	verify_results.py; verify_repeated_runs.py; rebuild_final_refinement.py	03_code (needs 04_evidence from the ZIP)
Public check of the headline numbers	scripts/verify_public_counts.py	GitHub
Project overview, claim–evidence ledger and reproduction steps	README.md; docs/CLAIMS.md; docs/REPRODUCING.md; docs/ARTEFACTS.md; docs/EXPERIMENT.md	GitHub
Question-level results	results/individual_methods.json; results/Independent_Repeat_Analysis.json	GitHub
Scoring rubric, August baseline sign-off record and decisions log	SCORING_RUBRICS.md; BASELINE_LOCK.md; decisions_log.md	02_documentation in the ZIP
Start page, results explorer, saved pages and answers	START_HERE.html; 04_evidence/Results_Explorer.html; 04_evidence/pages; 04_evidence/selected	ZIP

Appendix C. Final handover alignment — 25 October 2026

This revision corrects interpretation and delivery instructions in the supplied report. It preserves historical results and code. Both the professor assessment deck and the corrected FC leadership deck are included and clearly labelled. The leadership deck presents implementation recommendations, not a requirement for FC to repeat the sandbox.

What is established

Selected comparison: baseline 116/290 (40.0%); initial treatment 143/290 (49.3%); repeat 152/290 (52.4%). Repeat gain: 12.4 percentage points, approximately 31.0% relative. The independent whole-question bootstrap gives +5.2 to +20.0 points. Twenty-one question means improve, six decline and two are unchanged. Counts are observations, not guaranteed business effects.

Individual control: 114/290. Rendered content 133; grounded FAQ 134; structured facts 136; external facts 124; schema with visible structure 132; automatic rewrite 98; position diagnostic 167. The two intervals crossing zero are rendered content and external facts. Never add individual gains or substitute the individual control for the combined control.

The selected stack groups rendered content/cleanup, a joint question-and-facts block and FAQ restoration/repositioning. Its four technical operations are the same three content changes described in the leadership deck. Schema, external enrichment, generic rewrite and forced position are not added to this stack.

Matching establishes source identity/order and unchanged competitor text for the 290 headline pairs. It does not establish factual truth, unique use of FC content or citation entailment. The retained secondary reindexed result, 26/400 versus 27/400, does not establish a discovery gain. The reworded-query check is positive but inconclusive.

Controls were historical, captures differ, and development repeatedly used the evaluated questions. Human review of substantive citation support and calibrated relevance evaluation remain unresolved. No new paid generation was run for this handover.

Practical FC priorities from the leadership deck

Ownership and measurement. Assign engineering, content, commercial and analytics owners. Review available platform reporting, verified crawler requests and dated answer examples. Monitor citation inclusion separately from brand recommendation, referred traffic and customer enquiries.

Access and delivery. Preserve intended robots restrictions and check CDN/firewall behaviour. Browser captures on 25 October showed six dated Melbourne–London deal markers with JavaScript and none without it. This is sample evidence, not proof about every FC page or every engine. A spoofed user-agent or HTTP 200 is insufficient; inspect extracted facts and actual requests.

Useful content and structure. Make current offers, fees, conditions and FAQ questions readable in page text. Apply the labelled fact block where suitable. Generate appropriate structured data from the same approved record. Do not publish historical fares with a new validation date or model a generic London city route as a specific airport itinerary.

Offer integrity and customer pathways. Compare equivalent dates, cabins and conditions before calling a fare mismatch an error. Keep offers current and clarify booking, store and expert options. Genuine third-party coverage and accurate listings can support discovery, but our experiment did not quantify their causal effect.

Distribution options. Public crawling, search indexes, third-party references and direct app/API integrations are different routes. A connector decision needs its own product and commercial case; it is not conditional on repeating our fixed-source test.

Routine improvement. Publish through normal quality checks and maintain a reporting backlog. Observational monitoring does not itself prove causality. A confirmatory experiment is an optional future research activity, not a required FC implementation gate.

Sources: saved captures in 04_evidence/live_checks_20261025; Google AI optimisation guidance; Google reporting documentation; What Gets Cited (first citation in a two-source test, a different metric from ours). Reference details also appear in the FC deck.

Submission and reproduction map

Extract GEO_Final_Handover.zip completely, then open START_HERE.html. The private ZIP contains both labelled presentations, this report and agreement documentation, code snapshots, queries, captured corpus, saved page outputs and original answer records. The public GitHub companion contains the same current decks, report, selected code and numerical summaries; it intentionally excludes full third-party captures and answer archives.

Offline raw-record checks:
python 03_code/verify_results.py
python 03_code/verify_repeated_runs.py
The first script recomputes citations from saved answers and checks pairing/fingerprints. The second checks the retained initial/repeat comparison. Neither calls a model.

Public summary check:
python scripts/verify_public_counts.py
This validates aggregates, not raw answer scoring. It is not equivalent to the private verification.

Optional final-stage rebuild: install 03_code/refinement_requirements.txt and follow 03_code/README.md for rebuild_final_refinement.py. Live research generation requires provider credentials and original runtime configuration; costs and reproducibility limitations apply.

Presentation scope: 01_presentation/FC_GEO_Leadership_Final.pptx is for FC leadership. Professor_Final_With_Working_Demo.pptx is the retained assessment presentation. Earlier preparation scripts are labelled historical and are not the current FC speaking instructions.

Integrity: MANIFEST_SHA256.json records the private package files. HANDOVER_ALIGNMENT.json identifies shared public/private deliverables. Preserve historical run records and checksums. Updates to interpretation do not change original experiments.

Acceptance: delivery does not assert tutor/client approval, completion of independent human review, or verified personal contributions. Outstanding review is explicitly retained in the claim ledger and validation record.